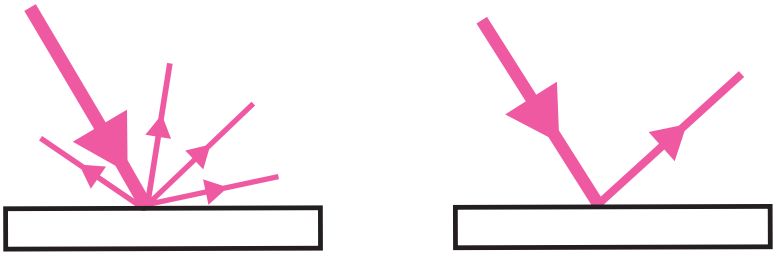
Reflection

When light bounces off something, we say it ______.

Light travels through the air in a __________.

Depending on the ______ light can reflect in two ways.
- in _______ reflection, light only leaves the surface in ___________.
- in ______ reflection, light leaves the surface in ___________.

1 The diagrams below show the two ways light can reflect.



(a) Which diagram shows **specular reflection**?

(b) Which diagram shows **diffuse reflection**?

2 Which kind of reflection is described in each case? Fill in the table.

Description	Specular or Diffuse?
Reflection in a mirror	
Light bouncing off a painted wall	
Reflection in a glass window	
Light bouncing off a friend's clothes	

3 If you want to see an image of yourself, which kind of reflection do you need?

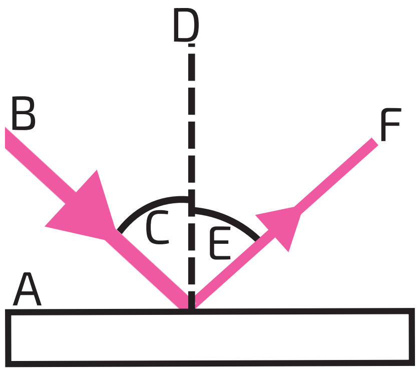
4 The brightness of reflected rays depend on the type of reflection. Complete the sentences using the words **weaker**, **energy**, **direction**, **specular**, **diffuse**.

Light from a ____________ reflection can be almost as bright as the incoming ray. This is because nearly all of the _________ of the ray is going the same way.

Light going in one direction from a ________ reflection will be much ________ than the incoming ray, because only a small fraction of the energy travels in that _________.

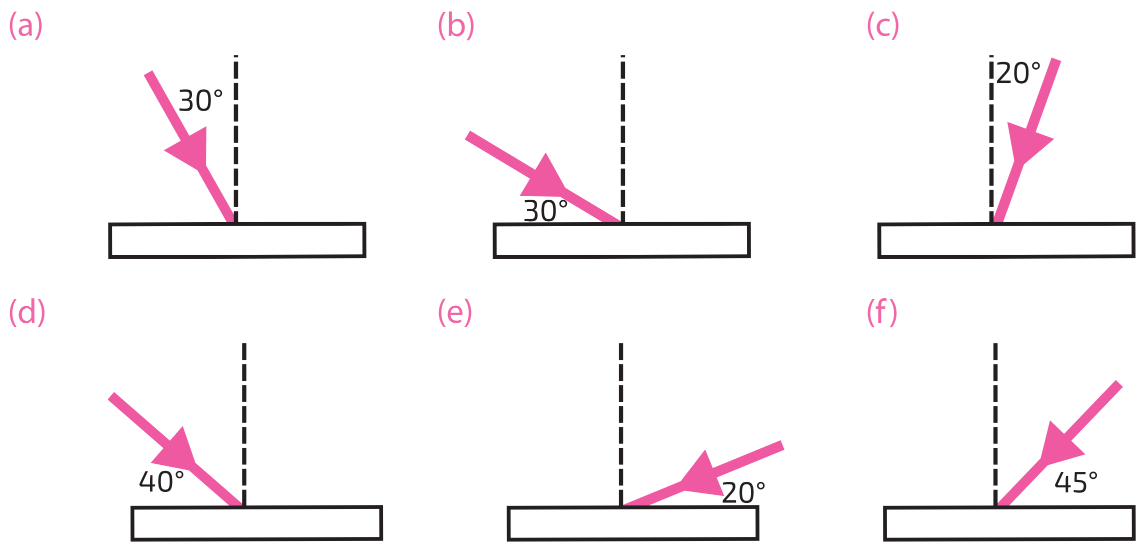
A line drawn at ____ angles to the surface is called the ______.

5 Which words describe the features in the diagram below? Fill in the table.

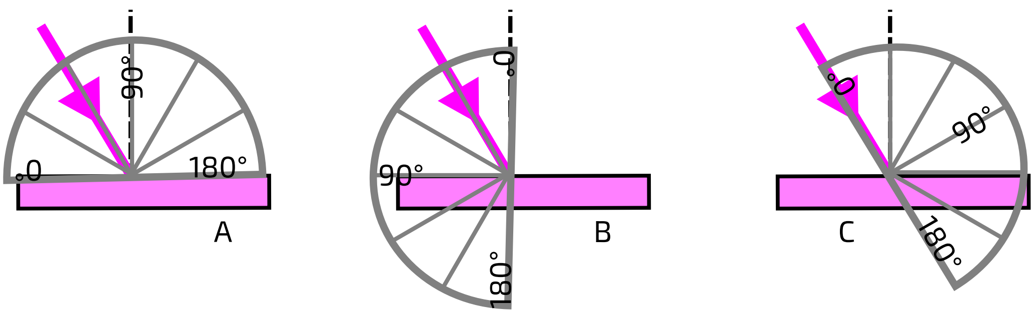


Technical term	A, B, C, D, E or F?
surface	
reflected ray	
incident ray	
angle of reflection	
angle of incidence	
normal	

6 In each diagram below, write down the angle of incidence.



7 The diagrams below show three attempts to measure an angle of incidence.

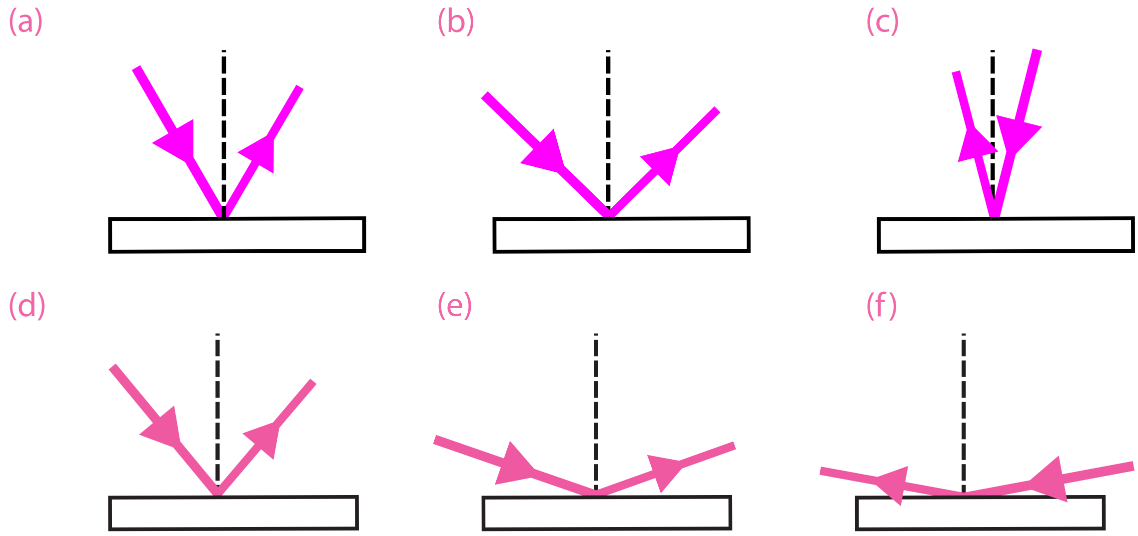


(a) From which of these protractors could we read off the correct angle?

(b) State the angle of incidence of the ray.

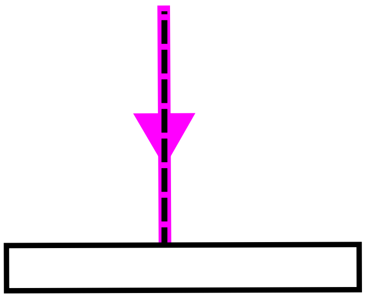
(c) What do we line the centre of the protractor up with?

8 Measure the angles of incidence and the angles of reflection using a protractor.



9 Look at your answers to question 8. How do you predict the angle of reflection if you are told the angle of incidence?

10 Look at the diagram below.

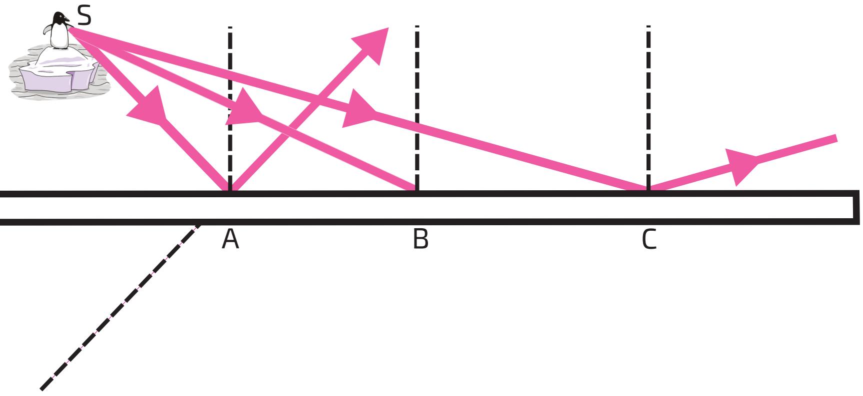


(a) State the angle of incidence.

(b) State what the angle of reflection will be.

(c) Draw the reflected ray on the diagram.

11 The diagram below shows three rays from the same point striking a mirror.



(a) Measure the angle of incidence of the ray going to B.

(b) Draw the reflected ray from B on the diagram using a protractor and ruler.

(c) Continue the reflected rays from B and C through the mirror with dashed lines. Use a ruler. The line from A has been done for you as an example.

(d) Where do the dashed lines meet? Describe the location in words.